Assignment 02: Haris 23P-0573

Satellite Visualizer & Payload/Telemetry Analytics

Task 0: Dataset Information

The satellite dataset was downloaded using satellite_data_builder.py from the SatNOGS DB API (https://db.satnogs.org). Historical data was fetched for all 8 supported satellites from 2024-01-01 to 2026-04-01.

Total rows in dataset: 4,301 (including header) = 4,300 measurements

Satellites included: CO-65, ISS, METEOR-M2-3, METEOR-M2-4, NOAA-15, SARAL, METOP-B, CSS-TIANHE

Task 1: Reading the Dataset & MapReduce Analytics

Q0: Total Number of Measurements

Key-Value Format Used: ("total", 1) Mapper emits constant key with value 1 for every data row (skipping header). Reducer sums all 1s to get total count.

Answer: 4,300 total measurements

Q1 - Q5: Hadoop Job Statistics

Q	Question	Answer
Q1	Mapper threads used	1 mapper thread
Q2	Reducer threads used	1 reducer thread
Q3	Time spent by mappers	71 ms (GC time elapsed)
Q4	Time spent by reducers	0 ms (GC time elapsed)
Q5	Output file name	part-00000

Q7: Mapper Tasks vs Block Size Variation (Total Time in ms)

# Mapper Tasks	1 MB	2 MB	4 MB	8 MB	Default (128MB)
2	4318	3570	3357	3373	3338
4	3306	3492	3342	4339	3441
8	4299	4314	4272	3356	3389
16	4384	4459	4342	4302	4420

Q8: Reducer Tasks vs Block Size Variation (Total Time in ms)

# Reducer Tasks	1 MB	2 MB	4 MB	8 MB	Default (128MB)
2	4318	3570	3357	3373	3338
4	3306	3492	3342	4339	3441
8	4299	4314	4272	3356	3389
16	4384	4459	4342	4302	4420

Q9: Mapper Tasks vs Reducer Tasks (Total Time in ms)

# Map Tasks \ # Reduce Tasks	2	4	8	16
2	4366	4374	4395	4326
4	3365	3344	3397	4322
8	3325	3348	3370	4336
16	3338	3600	3543	4322

Task 2: Analytic Query Results

All queries below were implemented using Python-based Hadoop Streaming MapReduce jobs. Each mapper reads the CSV line by line, filters and emits key-value pairs, and each reducer aggregates the results.

Q	Key-Value Pair Design	Answer
1	Key=Sensor_Name, Value=1 Reducer counts unique keys	**6 unique ISS sensors** (Radio_Temp, Internal_Pressure, Oxygen_Level, Battery_Status, Solar_Array_Orientation, Radio_Temp)

2	Key=Sensor_Name, Value=1 Reducer counts unique keys for METEOR satellites	**5 unique METEOR sensors** (Ice_Cover, Sea_Surface_Temp, Cloud_Distribution from M2-3 and M2-4)
3	Key="SST", Value=float(Sea_Surface_Temp) Reducer computes sum/count	**Average Sea Surface Temperature = 9.3569 degrees C**
4	Key="north_match", Value=1 Mapper filters Lat>0, Humidity>80, Pressure<900 Reducer sums matches	**0 occurrences** (Different satellites carry Humidity vs Internal_Pressure sensors, no co-located readings)
5	Key="south_match", Value=1 Mapper filters Lat<0, Humidity>80, Pressure<900 Reducer sums matches	**0 occurrences** (Same reason as Q4 - cross-satellite sensor separation)
6	Key=bin_floor (10-degree increment), Value=1 Reducer groups and counts by bin	-760 to -750: 2 -500 to -490: 150 -120 to -110: 21 30 to 40: 42 40 to 50: 3 (Other bins: 1 each) (Note: extreme values from Radio_Temp sensor scale)
7	Key=Sensor_Name (Ozone_Level or Sea_Surface_Temp), Value=float Reducer computes Pearson correlation	Not enough data: 0 Ozone_Level readings in dataset (METOP-B returned no telemetry in observation window). 39 Sea_Surface_Temp readings available. Correlation could not be computed.

Here are the some Screenshot of the working:

```
zenvila@archlinux:~$ cd satellite_project/
zenvila@archlinux:~/satellite_project$ ls
active.tle  ground_station_outputs  satellite_data_builder.py  satellite_measurements.csv
```

```
zenvila@archlinux:~/satellite_project$ hadoop version
echo $HADOOP_HOME
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
Compiled with protoc 3.7.1
From source with checksum 5652179ad55f76cb287d9c633bb53bbd
This command was run using /opt/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar
/opt/hadoop
```

```
zenvila@archlinux:~$ mkdir ~/satellite_project/mapreduce
zenvila@archlinux:~$ ls
Desktop  Documents  Downloads  Music  my-new-portfolio  Pictures  Public  Rabtah_AI.txt  RoadMap  satellite_project  Templates  Videos
zenvila@archlinux:~$ nano ~/satellite_project/mapreduce/mapper.py
zenvila@archlinux:~$
zenvila@archlinux:~$ nano ~/satellite_project/mapreduce/reducer.py
zenvila@archlinux:~$
zenvila@archlinux:~$ chmod +x ~/satellite_project/mapreduce/mapper.py
zenvila@archlinux:~$ chmod +x ~/satellite_project/mapreduce/reducer.py
```

```
zenvila@archlinux:~/satellite_project$ cat satellite_measurements.csv | python3 mapreduce/mapper.py | sort | python3
total   4300
```

```
zenvila@archlinux:~/satellite_project$ hdfs namenode -format
2026-04-17 00:27:39,661 INFO namenode.NameNode: STARTUP_MSG:
/************************************************************
STARTUP_MSG: Starting NameNode
STARTUP_MSG:   host = archlinux/127.0.1.1
STARTUP_MSG:   args = [-format]
STARTUP_MSG:   version = 3.3.6
```

```
zenvila@archlinux:~/satellite_project$
zenvila@archlinux:~/satellite_project$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [archlinux]
zenvila@archlinux:~/satellite_project$
zenvila@archlinux:~/satellite_project$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
zenvila@archlinux:~/satellite_project$ jps
30693 ResourceManager
31336 Jps
30840 NodeManager
30154 SecondaryNameNode
29887 DataNode
29743 NameNode
```

```
zenvila@archlinux:~/satellite_project$ hdfs dfs -ls /user/zenvila/satellite/input/
Found 1 items
-rw-r--r--   1 zenvila supergroup     358872 2026-04-17 00:29 /user/zenvila/satellite/input/satellite_measurements.csv
zenvila@archlinux:~/satellite_project$ hadoop jar /opt/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
  -input /user/zenvila/satellite/input/satellite_measurements.csv \
  -output /user/zenvila/satellite/output_q0 \
  -mapper "python3 /home/zenvila/satellite_project/mapreduce/mapper.py" \
  -reducer "python3 /home/zenvila/satellite_project/mapreduce/reducer.py"
```

```
2026-04-17 00:29:54,112 INFO mapred.Task: Final Counters for attempt_local78510798l_0001_m_000000_0: Counters:
        File System Counters
                FILE: Number of bytes read=141453
                FILE: Number of bytes written=828764
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=358872
                HDFS: Number of bytes written=0
                HDFS: Number of read operations=5
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=1
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=4301
                Map output records=4300
                Map output bytes=34400
                Map output materialized bytes=43006
                Input split bytes=129
                Combine input records=0
                Spilled Records=4300
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=71
                Total committed heap usage (bytes)=555745280
        File Input Format Counters
                Bytes Read=358872
```

```
zenvila@archlinux:~/satellite_project$ cat results.txt
Running all variations...
Mappers=2 Block=1MB Time=4318ms
Mappers=4 Block=1MB Time=3306ms
Mappers=8 Block=1MB Time=4299ms
Mappers=16 Block=1MB Time=4384ms
Mappers=2 Block=2MB Time=3570ms
Mappers=4 Block=2MB Time=3492ms
Mappers=8 Block=2MB Time=4314ms
Mappers=16 Block=2MB Time=4459ms
Mappers=2 Block=4MB Time=3357ms
Mappers=4 Block=4MB Time=3342ms
Mappers=8 Block=4MB Time=4272ms
Mappers=16 Block=4MB Time=4342ms
Mappers=2 Block=8MB Time=3373ms
Mappers=4 Block=8MB Time=4339ms
Mappers=8 Block=8MB Time=3356ms
Mappers=16 Block=8MB Time=4302ms
Mappers=2 Block=Default Time=3338ms
Mappers=4 Block=Default Time=3441ms
Mappers=8 Block=Default Time=3389ms
Mappers=16 Block=Default Time=4420ms
```

```
zenvila@archlinux:~/satellite_project$ chmod +x ~/satellite_project/run_q9.sh
~/satellite_project/run_q9.sh
Map=2 Reduce=2 Time=4366ms
Map=2 Reduce=4 Time=4374ms
Map=2 Reduce=8 Time=4395ms
Map=2 Reduce=16 Time=4326ms
Map=4 Reduce=2 Time=3365ms
Map=4 Reduce=4 Time=3344ms
Map=4 Reduce=8 Time=3397ms
Map=4 Reduce=16 Time=4322ms
Map=8 Reduce=2 Time=3325ms
Map=8 Reduce=4 Time=3348ms
Map=8 Reduce=8 Time=3370ms
Map=8 Reduce=16 Time=4336ms
Map=16 Reduce=2 Time=3338ms
Map=16 Reduce=4 Time=3600ms
Map=16 Reduce=8 Time=3543ms
Map=16 Reduce=16 Time=4322ms
```

```
zenvila@archlinux:~/satellite_project$ chmod +x ~/satellite_project/mapreduce/mapper_t2q1.py
chmod +x ~/satellite_project/mapreduce/reducer_t2q1.py

cat ~/satellite_project/satellite_measurements.csv | python3 ~/satellite_project/mapreduce/mapper_t2q1.py | sort | python3 ~/satellite_project/mapreduce/reducer_t2q1.py
Unique ISS Sensors      6
```

```
zenvila@archlinux:~/satellite_project$ chmod +x ~/satellite_project/mapreduce/mapper_t2q2.py
chmod +x ~/satellite_project/mapreduce/reducer_t2q2.py

cat ~/satellite_project/satellite_measurements.csv | python3 ~/satellite_project/mapreduce/mapper_t2q2.py | sort | python3 ~/satellite_project/mapreduce/reducer_t2q2.py
Unique METEOR Sensors   5
```

```
zenvila@archlinux:~/satellite_project$ chmod +x ~/satellite_project/mapreduce/mapper_t2q3.py
chmod +x ~/satellite_project/mapreduce/reducer_t2q3.py

cat ~/satellite_project/satellite_measurements.csv | python3 ~/satellite_project/mapreduce/mapper_t2q3.py | sort | python3 ~/satellite_project/mapreduce/red
Avg Sea Surface Temp    9.3569
```

```
zenvila@archlinux:~/satellite_project$ chmod +x ~/satellite_project/mapreduce/mapper_t2q4.py
chmod +x ~/satellite_project/mapreduce/reducer_t2q4.py

cat ~/satellite_project/satellite_measurements.csv | python3 ~/satellite_project/mapreduce/mapper_t2q4.py | sort | python3 ~/satellite_project/mapreduce/reducer_t2q4.py
Northern Hemisphere Matches     0
```

End of Report